

## 09/18 Advanced Statistics HW

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**Q1:**  $A(\eta) = -n \ln p_0 = n \ln \left[ 1 + \sum_{i=1}^s (e^{\eta_i}) \right]$

Since  $X_i \sim \text{Multinomial}(n; p_1, p_2, \dots, p_s)$ , it follows that

$$p(X = x) = \frac{n!}{\prod_{i=0}^s x_i!} \prod_{i=0}^s p_i^{x_i}$$

and  $\sum_{i=0}^s x_i = n$ ,  $\sum_{i=0}^s p_i = 1$ . Moreover, we define  $\eta_i = \ln \frac{p_i}{p_0}$ , then the pmf can be written as

$$p(X = x) = \frac{n!}{\prod_{i=0}^s x_i!} \prod_{i=0}^s p_i^{x_i} = \frac{n!}{\prod_{i=0}^s x_i!} p_0^{x_0} \prod_{i=1}^s p_i^{x_i} = \frac{n!}{\prod_{i=0}^s x_i!} p_0^{n - \sum_{i=1}^s x_i} \prod_{i=1}^s (p_0 e^{\eta_i})^{x_i},$$

where

$$p_0^{n - \sum_{i=1}^s x_i} = \exp \left\{ \left( n - \sum_{i=1}^s x_i \right) \ln p_0 \right\} = \exp \left\{ n \ln p_0 - \left( \sum_{i=1}^s x_i \right) \ln p_0 \right\},$$

$$\prod_{i=1}^s (p_0 e^{\eta_i})^{x_i} = \exp \left\{ \sum_{i=1}^s \eta_i x_i + \left( \sum_{i=1}^s x_i \right) \ln p_0 \right\}.$$

Then we have

$$p(X = x) = \frac{n!}{\underbrace{\prod_{i=0}^s x_i!}_{h(x)}} \exp \left\{ \sum_{i=1}^s \underbrace{\eta_i x_i}_{\eta_i T_i(x)} + \underbrace{n \ln p_0}_{-A(\eta)} \right\}$$

Therefor,  $A(\eta) = -n \ln p_0 = -n \ln \frac{1}{1 + \sum_{i=1}^s e^{\eta_i}} = n \ln(1 + \sum_{i=1}^s e^{\eta_i})$  where  $\eta_i = \ln \frac{p_i}{p_0} \implies p_i = p_0 e^{\eta_i} \implies p_0 + \sum_{i=1}^s p_0 e^{\eta_i} = 1 \implies p_0 = (1 + \sum_{i=1}^s e^{\eta_i})^{-1}$

**Q2:**  $X_i \stackrel{iid}{\sim} N(\mu, \sigma^2)$ ,  $f_X(x)$ : joint density belongs to exponential family, Natural parameter  $\eta = ?$  Natural parameter space ?  $T(x) = ?$

**Q3:**  $E(X_i) = np_i, Cov(X_i, X_j) = \begin{cases} np_i(1 - p_i) & \text{if } i = j \\ -np_i p_j & \text{if } i \neq j \end{cases}$

Since Q1, we know that  $A(\eta) = n \ln(1 + \sum_{i=1}^s e^{\eta_i})$  and  $p_i = \frac{e^{\eta_i}}{1 + \sum_{i=1}^s e^{\eta_i}}$

$$E[X_i] = \frac{\partial A}{\partial \eta_i} = n \frac{e^{\eta_i}}{1 + \sum_{i=1}^s e^{\eta_i}} = np_i$$

$$Cov(X_i, X_j) = \frac{\partial^2 A}{\partial \eta_i \partial \eta_j} = \text{two cases about } i \neq j \text{ and } i = j$$

for  $i \neq j$